

As well-known, $(\mathbb{R}, +, \cdot)$ is the Complete ordered field.
Consider a set $\mathbb{R}^2$. Define two operations on $\mathbb{R}^2$ as:

$$(x_1, y_1) + (x_2, y_2) \stackrel{\text{def}}{=} (x_1 + x_2, y_1 + y_2)$$

$$(x_1, y_1) \cdot (x_2, y_2) \stackrel{\text{def}}{=} (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2)$$

operation on $\mathbb{R}^2$

defined on $\mathbb{R}$.

$$(x, y)^{-1} =$$

Then, the triple $(\mathbb{R}^2, +, \cdot)$ is a field:

$$\begin{cases} \text{Identity} & (0, 0) \\ \text{Unity} & (1, 0) \\ \text{inverse} & (x, y)^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right) \end{cases}$$

Def. Define a Norm of $(a, b) \in \mathbb{C}$ as:

$$\|(a, b)\| \stackrel{\text{def}}{=} \sqrt{a^2 + b^2}.$$

Lemma. $\|z\|^2 = z \cdot \bar{z}$. Moreover, if $z \neq 0$, then $z^{-1} = \frac{\bar{z}}{\|z\|^2}$

$$\begin{aligned} (i h) \cdot (-i h) \\ = -i^2 \cdot h^2 = h^2. \end{aligned}$$

Consider a Complex map:

$$f: \Omega \rightarrow \mathbb{C} : (x, y) \mapsto (u(x, y), v(x, y))$$

Where $\Omega \subseteq \mathbb{C}$. Set-Theoretically, it equals to a map from a subset of $\mathbb{R}^2$ into $\mathbb{R}^2$.
Moreover, Topologically, $\mathbb{C}$ is a Metric space induced by $\|\cdot\|$, homeomorphic to $\mathbb{R}^2$.

Therefore, the continuity of f is not different whether we see that $\mathbb{C}$ as $\mathbb{C}$ or $\mathbb{R}^2$.
But, in the Differentiation, Complex and $\mathbb{R}^2$ have crucial difference. Recall that

$f: A(\subset \mathbb{R}^m) \rightarrow \mathbb{R}^n$ is differentiable at $a \in A^\circ$ if and only if there exists a linear $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ s.t.

$$\frac{f(a+h) - f(a) - T(h)}{\|h\|_m} \rightarrow 0 \text{ as } h \rightarrow 0.$$

If the derivative T of f at $a \in A^\circ$ exists, it is unique.

Equivalently, if f has a derivative T at $a \in A^\circ$, then for small enough $h \in \mathbb{R}^m$,

$$f(a+h) - f(a) = \|h\| \cdot \left[\frac{f(a+h) - f(a) - T(h)}{\|h\|} \right] + T(h)$$

Moreover, if f is differentiable at $a \in A^\circ$, then all directional derivatives of f at a exist,

$$f'(a)u = Df(a) \cdot u$$

But, since $\mathbb{C}$ is a field, we can define derivative without a norm,

this is the first bifurcation that indicates $\mathbb{C}$ does not equal $\mathbb{R}^2$.

Def: Let $\Omega \subseteq \mathbb{C}$ open, and let $f: \Omega \rightarrow \mathbb{C}$ be a map.
Given $z_0 \in \Omega$, f is said to be holomorphic at z_0 if: the limit

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} \quad (h \in \mathbb{C}, h \neq 0)$$

exists. The limit is called a derivative of f at z_0 , denote $f'(z_0)$.

Note: Holomorphic = Regular = Complex differentiable $\stackrel{\text{not definition}}{=} \text{Analytic}$

Cauchy - Riemann equation.

Let $f: \Omega \rightarrow \mathbb{C}: (x, y) \mapsto (u(x, y), v(x, y))$ be a map where $\Omega \subseteq \mathbb{C}$ open set.

If f is holomorphic at $z_0 = (x_0, y_0)$, then the partial derivatives of u, v at z_0 exist and

$$\frac{\partial u}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) \quad \text{and} \quad \frac{\partial u}{\partial y}(z_0) = -\frac{\partial v}{\partial x}(z_0) \quad \dots (*)$$

hold. (Or $u_x(z_0) = v_y(z_0)$ and $u_y(z_0) = -v_x(z_0)$)

Conversely, if u and v are continuously differentiable at $z_0 = (x_0, y_0)$ satisfying (*), then f is holomorphic at z_0 . More specifically,

$$f'(z_0) = \frac{\partial u}{\partial x}(z_0) + i \cdot \frac{\partial v}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) - i \cdot \frac{\partial u}{\partial y}(z_0)$$

Proof. Suppose that f is holomorphic at z_0 . Then,

$$\begin{aligned} f'(z_0) &= \lim_{h_1 \rightarrow 0} \frac{f(x_0 + h_1, y_0) - f(x_0, y_0)}{h_1} = \lim_{h_2 \rightarrow 0} \frac{f(x_0, y_0 + h_2) - f(x_0, y_0)}{ih_2} \\ &= \frac{\partial f}{\partial x}(z_0) = \frac{1}{i} \cdot \frac{\partial f}{\partial y}(z_0) \end{aligned}$$

Therefore, at z_0 , $u_x + i \cdot v_x = v_y - i \cdot u_y$.

Conversely, suppose that u, v are continuously differentiable at z_0 . That is,

u_x, u_y, v_x, v_y exist and are continuous at z_0 .

Thm. Define a complex map as

$$f: \mathbb{D}_R \rightarrow \mathbb{C} : z \mapsto \sum_{n=0}^{\infty} a_n z^n$$

where $R = \left[\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} \right]^{-1}$. Then, f is differentiable by terms on the domain as

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

Proof. Since $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$, we obtain

$$\limsup |a_n|^{\frac{1}{n}} = \limsup |n a_n|^{\frac{1}{n}}$$

Thus $\sum a_n z^n$ and $\sum n a_n z^{n-1}$ have same convergence boundary,

Define a map on $\mathbb{D}_R$ as:

$$g(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

Enough to show that $g = f'$. Let $z_0 \in \mathbb{D}_R$ be fixed, and $r \in \mathbb{R}$ s.t. $|z_0| < r < R$.

Since $f(z)$ converges, denote $S_N(z) = \sum_{n=0}^N a_n z^n$ and $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$. ($f = S_N + E_N$).

For $h \in \mathbb{C}$ s.t. $|z_0 + h| < r$,

$$\frac{f(z_0 + h) - f(z_0)}{h} - g(z_0) = \left[\frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right] + (S'_N(z_0) - g(z_0)) + \left[\frac{E_N(z_0 + h) - E_N(z_0)}{h} \right]$$

$$\text{Meanwhile, } \left| \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right| \leq \sum_{n=N+1}^{\infty} |a_n| \cdot \left| \frac{(z_0 + h)^n - z_0^n}{h} \right| \leq \sum_{n=N+1}^{\infty} |a_n| \cdot n r^{n-1} \rightarrow 0 \text{ as } N \rightarrow \infty$$

$$(z_0 + h)^n - z_0^n = (z_0 + h - z_0) \left[(z_0 + h)^{n-1} + (z_0 + h)^{n-2} z_0 + \dots + z_0^{n-1} \right]$$

And clearly $S'_N(z_0) - g(z_0) \rightarrow 0$ as $N \rightarrow \infty$. And,

$$\left| \frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right| \rightarrow 0 \text{ as } h \rightarrow 0.$$

Therefore $g = f'$ for any $z_0 \in \mathbb{D}_R$.

Corollary. Power series defined on convergence disk is C^∞ .

Curve.

A continuous map $Z: [a, b] \rightarrow \mathbb{C}; t \mapsto Z(t)$ is called a Parametrized Curve.

If it is 'injection', then called Simple,

If $Z(a) = Z(b)$, then called closed.

It is called simple closed if injective on $[a, b)$.

Moreover, if Z is continuously differentiable on $[a, b]$ and $Z' \neq 0$, then it is smooth.

A smooth curves $\begin{cases} Z: [a, b] \rightarrow \mathbb{C} \\ \tilde{Z}: [c, d] \rightarrow \mathbb{C} \end{cases}$ are equivalent if: there is $t: [c, d] \rightarrow [a, b]$ s.t.
 t is 1-1, t is continuously differentiable, $t' > 0$, $\tilde{Z} = Z \circ t$.

Integral

Def. Let γ be a smooth curve parametrized by $Z: [a, b] \rightarrow \mathbb{C}$, and let $f: \mathbb{C} \rightarrow \mathbb{C}$ be a continuous map on γ . Define an Integral of f along γ as:

$$\int_{\gamma} f(z) dz \stackrel{\text{def}}{=} \int_a^b f(Z(t)) \cdot Z'(t) dt.$$

And $\text{length}(\gamma) = \int_a^b |Z'(t)| dt.$

Thm. Let $f: \Omega \rightarrow \mathbb{C}$ be a map where $\Omega \subseteq \mathbb{C}$ is open, and γ be a curve from w_1 to w_2 . If f has primitive function F on Ω , then

$$\int_{\gamma} f(z) dz = F(w_2) - F(w_1).$$

Goursat's Thm. Let $f: \Omega \rightarrow \mathbb{C}$ be a holomorphic and a triangle $T \subseteq \Omega$. Then,

$$\int_T f(z) dz = 0.$$

Lemma. In open disk, a holomorphic map has a primitive map in the disk.

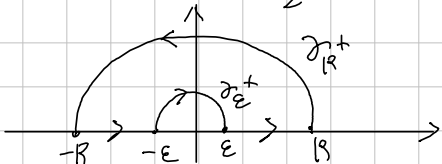
Corollary. If f is holomorphic in the disk, then for any closed curve γ in the disk,

$$\int_{\gamma} f(z) dz = 0.$$

Calculate

prove $\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$.

Proof. Let $f(z) = \frac{1 - e^{iz}}{z^2}$. And, set the contour as



Observe that: for any $0 < \epsilon < R < \infty$,

$$\int_{\gamma} f(z) dz = \int_{-R}^{-\epsilon} \frac{1 - e^{iz}}{z^2} dx + \int_{\gamma_{\epsilon}^+} \frac{1 - e^{iz}}{z^2} dz + \int_{\epsilon}^R \frac{1 - e^{iz}}{z^2} dx + \int_{\gamma_R^+} \frac{1 - e^{iz}}{z^2} dz = 0.$$

First, $\left| \frac{1 - e^{iz}}{z^2} \right| \leq \frac{2}{|z|^2}$ implies $\left| \int_{\gamma} f(z) dz \right| \leq \sup_{z \in \gamma_R^+} \left| \frac{1}{z^2} \right| \cdot \text{length } \gamma_R^+ = \frac{1}{R^2} \cdot \pi R \rightarrow 0$

as $R \rightarrow \infty$. Therefore, $\int_{|x| \geq \epsilon} \frac{1 - e^{ix}}{x^2} dx = - \int_{\gamma_{\epsilon}^+} \frac{1 - e^{iz}}{z^2} dz$.

Since $\gamma_{\epsilon}^-: z(t) = \epsilon \cdot e^{it}$ ($t \in [0, \pi]$), and observe that

$$\frac{1 - e^{iz}}{z^2} = \frac{-zi + zi + 1 - e^{iz}}{z^2} = -\frac{i}{z} + \frac{1 + zi - e^{iz}}{z}$$

By L'Hopital Thm, $\lim_{z \rightarrow 0} \frac{1 + zi - e^{iz}}{z} = \lim_{z \rightarrow 0} i + ie^{iz} = i + i = 2i$.

Therefore $\frac{1 + zi - e^{iz}}{z}$ is bounded in the neighborhood of zero. Take $M > 0$ as bound.

Now, $\int_{\gamma_{\epsilon}^-} -\frac{i}{z} dz = \int_0^{\pi} \frac{i}{\epsilon \cdot e^{it}} \cdot \epsilon \cdot i \cdot e^{it} dt = \int_0^{\pi} 1 dt = \pi$

and $\int_{\gamma_{\epsilon}^-} \frac{1 + zi - e^{iz}}{z} dz \leq M \cdot \pi \epsilon \rightarrow 0$ as $\epsilon \rightarrow 0$

Finally, $\int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} dx = \lim_{\epsilon \rightarrow 0} \int_{|x| \geq \epsilon} f(z) dz = \pi$.

Since $\frac{1 - \cos x}{x^2}$ is even, $\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$.

Thm. Cauchy Integral Formula.

Suppose f is Holomorphic on some open set containing a closure of disk D .
If C is a Curve with Positive orientation of ∂D , then for any $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0} dz$$

Proof. Let $z_0 \in D$ be fixed, and consider a Curve Γ_{ϵ} as
Since $\frac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, therefore



$$\int_{\Gamma_{\epsilon}} \frac{f(z)}{z - z_0} dz = 0. \text{ Taking } \epsilon \rightarrow 0, \text{ then we obtain}$$

$$\lim_{\epsilon \rightarrow 0} \int_{\Gamma_{\epsilon}} \frac{f(z)}{z - z_0} dz = \int_{C_{\epsilon}} \frac{f(z)}{z - z_0} dz + \int_C \frac{f(z)}{z - z_0} dz = 0.$$

Meanwhile, $\frac{f(z)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} + \frac{f(z_0)}{z - z_0}$ and since f is Holomorphic,
the first term is bounded, therefore $\lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{f(z) - f(z_0)}{z - z_0} dz = 0.$

$$\begin{aligned} \text{Now, } \lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{f(z_0)}{z - z_0} dz &= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{1}{z - z_0} dz \\ &= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \left(\int_0^{2\pi} \frac{1}{\epsilon \cdot e^{it}} \cdot \epsilon \cdot i \cdot e^{it} dt \right) \\ &= -f(z_0) \cdot 2\pi i. \end{aligned}$$

$$\text{Consequently, } \int_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

Corollary. $\int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}.$

Proof. Using induction for n . For $n=0$, clear. If $f^{(n-1)}(z_0) = \frac{(n-1)!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^n} dz$, then

$$\begin{aligned} \frac{f^{(n-1)}(z_0 + h) - f^{(n-1)}(z_0)}{h} &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[\frac{1}{(z - z_0 - h)^n} - \frac{1}{(z - z_0)^n} \right] dz \\ &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[\frac{h}{(z - z_0 - h)(z - z_0)^n} \cdot \left(\sum_{k=0}^{n-1} \frac{1}{(z - z_0 - h)^{n-1-k}} \cdot \frac{1}{(z - z_0)^k} \right) \right] dz \\ &\rightarrow \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{(z - z_0)^2} \cdot \sum_{k=0}^{n-1} \frac{1}{(z - z_0)^{n-1-k}} dz \quad \text{as } h \rightarrow 0 \\ &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot n \cdot \frac{1}{(z - z_0)^{n+1}} dz \\ &= \frac{n!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = f^{(n)}(z_0). \end{aligned}$$

Thm. Cauchy inequality.

$$|f^{(n)}(z_0)| \leq \frac{n!}{R} \cdot \sup_C |f| \text{ if } f \text{ is Holomorphic in an open containing } \overline{B_R(z_0)}, C = \partial B_R(z_0).$$

Cauchy's Integral Formula, Revised.

Thm. Let $f: G \rightarrow \mathbb{C}$ be a Holomorphic on open $G \subset \mathbb{C}$.

If $z_0 \in G$ and $\overline{B_r(z_0)} \subseteq G$, then for a Curve $C: z_0 + re^{it}$ ($0 \leq t \leq 2\pi$),

$$f(z_0) = \frac{1}{2\pi i} \cdot \oint_C \frac{f(z)}{z - z_0} dz.$$

Proof. Denote $D = B_r(z_0)$. Let $z_0 \in D$ be fixed, and consider a Curve $\gamma_{\delta, \epsilon}$ as

Since $\frac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, $\int_{\gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz = 0$.



Taking $\delta \rightarrow 0$, then we obtain

$$\lim_{\delta \rightarrow 0} \int_{\gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz = - \int_{C_\epsilon} \frac{f(z)}{z - z_0} dz + \int_C \frac{f(z)}{z - z_0} dz = 0.$$

Meanwhile, $\frac{f(z)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} + \frac{f(z_0)}{z - z_0}$ and since f is Holomorphic,

the first term is bounded, therefore $\lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{f(z) - f(z_0)}{z - z_0} dz = 0$.

$$\text{Now, } \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{f(z_0)}{z - z_0} dz = f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{1}{z - z_0} dz$$

$$= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \left(\int_0^{2\pi} \frac{1}{\epsilon \cdot e^{i\epsilon}} \cdot \epsilon \cdot i \cdot e^{i\epsilon} dt \right)$$

$$= f(z_0) \cdot 2\pi i.$$

$$\int_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

Consequently,

$$\int_C \frac{f(z)}{z - z_0} dz = \lim_{\delta \rightarrow 0} \int_{\gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz + \int_{C_\epsilon} \frac{f(z)}{z - z_0} dz = 2\pi i \cdot f(z_0).$$

$$\text{Corollary. } \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = 2\pi i \frac{f^{(n)}(z_0)}{n!}$$

Thm. Suppose f is Holomorphic on an open G containing a closed disk $\overline{D}$ having center z_0 ,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n \quad (z \in D)$$

Proof.
$$f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} d\mu \quad \left(\frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\mu - z_0}\right)} \right)$$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z - z_0)^n d\mu$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n$$

Conclary, Liouville's Theorem.

If $f: \mathbb{C} \rightarrow \mathbb{C}$ is entire and bounded, then it is constant.

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$, $|f'(z_0)| \leq \frac{1}{R} \cdot \sup_{C_R} |f| \rightarrow 0$ as $R \rightarrow \infty$. Therefore, $f' = 0$.
 Thus f is constant.

Fundamental Theorem of Algebra.

Non-constant polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire. observe that
$$\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right) \quad (z \neq 0).$$

The terms $a_{n-k} \cdot z^{-k}$ ($k=1, \dots, n$) converges to zero as $|z| \rightarrow \infty$, thus for some $R > 0$,

$$\begin{aligned} |P(z)| &\geq 2|a_n| \cdot z^n \quad (|z| > R) \\ &\geq 2|a_n| \cdot R^n \end{aligned}$$

Therefore it is bounded below in $|z| > R$, and clearly in $|z| \leq R$.

Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.

Thm. Let G be a connected open set and let $f: G \rightarrow \mathbb{C}$ be an analytic function. Then, TFAE:

a) $f \equiv 0$.

b) There is a point $a \in G$ such that $f^{(n)}(a) = 0$ for each $n \geq 0$.

c) $f^{-1}[\{0\}]$ has a limit point in G .

Proof. a) $\Rightarrow$ b) and a) $\Rightarrow$ c) are trivial. To show c) $\Rightarrow$ a),

let $a \in G$ be a limit point of $Z = f^{-1}[\{0\}]$. Since Z is closed, $a \in Z$, or $f(a) = 0$.

Suppose that for some $n \in \mathbb{N}$,

$$f(a) = f'(a) = \dots = f^{(n-1)}(a) = 0 \text{ but } f^{(n)}(a) \neq 0.$$

Since f is analytic,
$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (z-a)^k = \sum_{k=n}^{\infty} \frac{f^{(k)}(a)}{k!} (z-a)^k.$$

in $|z-a| < R$, where $R > 0$ s.t. $B_R(a) \subset G$.

Put
$$g(z) = \sum_{k=n}^{\infty} a_k (z-a)^{k-n} \quad \left(a_k = \frac{f^{(k)}(a)}{k!} \right)$$

then, g is analytic in $B_R(a)$, and $f = (z-a)^n g$, and $g(a) = a_n \neq 0$.

Now, we can find $0 < r < R$ s.t. $g(z) \neq 0$ for $|z-a| < r$, but since a is the limit point of Z , so for some b s.t. $f(b) = 0$ and $0 < |b-a| < r$, thus

$$f(b) = 0 = (b-a)^n \cdot g(b) \Rightarrow g(b) = 0,$$

which is Contradiction.

⌈ Singularities.

⌈ Def. A complex function f has an isolated singularity at $z=a$ if:

there is an $R > 0$ st f is defined and analytic in $B_R(a) - \{a\}$, but not in $B_R(a)$.

An isolated singularity $a \in \mathbb{C}$ of f is called:

- Removable if: there is an analytic extension of f to $B_R(a)$.
- Pole if: $\lim_{z \rightarrow a} |f(z)| = \infty$.
- essential if: neither Removable nor Pole.

⌈ Thm. Let $a \in \mathbb{C}$ be an isolated singularity of f . Then,

$a \in \mathbb{C}$ is removable singularity iff $\lim_{z \rightarrow a} (z-a) \cdot f(z) = 0$.

⌈ Prop. If $G \subseteq \mathbb{C}$ is a region containing $a \in \mathbb{C}$ and f is analytic on $G \setminus \{a\}$ with a pole a , then there exists $m \in \mathbb{N}$ and analytic $g: G \rightarrow \mathbb{C}$ such that

$$f(z) = \frac{g(z)}{(z-a)^m}$$

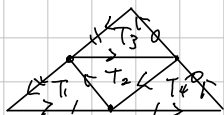
The smallest such a number is called order of a pole.



Cauchy - Goursat's Thm.

Thm. A map f is holomorphic on open set $\Omega \subseteq \mathbb{C}$ and a triangle T is contained in Ω then $\int_T f(z) dz = 0$.

Proof. Let $T^{(0)} = T$ with positive oriented, and $T_1^{(1)}, T_2^{(1)}, T_3^{(1)}, T_4^{(1)}$ be:



Then, $\int_{T^{(0)}} f(z) dz = \sum_{i=1}^4 \int_{T_i^{(1)}} f(z) dz$, because the integral on T_2 is canceled.

So, for some $j \in \{1, 3, 4\}$, $\left| \int_{T^{(0)}} f(z) dz \right| \leq 4 \cdot \left| \int_{T_j^{(1)}} f(z) dz \right|$

Put $T^{(0)} = T_j^{(1)}$. Inductively, for each $n \in \mathbb{N}$,

$$\left| \int_{T^{(n)}} f(z) dz \right| \leq 4^n \cdot \left| \int_{T^{(0)}} f(z) dz \right|$$

And, by construction,

$$T^{(0)} \supseteq T^{(1)} \supseteq \dots$$

Def. Let G be an open set of $\mathbb{C}$.

A function $f: G \rightarrow \mathbb{C}$ is called Analytic if f is continuously differentiable.

Logarithm

Def. If $G \subseteq \mathbb{C}$ is an open connected set in $\mathbb{C}$ and $f: G \rightarrow \mathbb{C}$ is a continuous function s.t.
 $z = \exp f(z)$ for all $z \in G$

Then, f is called a branch of the logarithm.

Explicitly, if $z = x + iy \in G$, then let $\log z = a + ib$. Now,

$$\exp(\log z) = \exp(a) \cdot \exp(ib) = e^a \cdot (\cos b + i \sin b) = x + iy.$$

Thus, $\begin{cases} e^a \cos b = x \\ e^a \sin b = y \end{cases}, \quad \begin{cases} |e^a (\cos b + i \sin b)| = |e^a| = |z|, \text{ so } a = \ln |z|. \\ b = \arg z. \end{cases}$

Therefore, $\log z = \ln |z| + i \arg z$.

Cauchy-Riemann equation.

Let $f: G \rightarrow \mathbb{C}$ be a map on region G and let $U(x,y) = \operatorname{Re} f(x,y)$, $V(x,y) = \operatorname{Im} f(x,y)$. Suppose that U and V have continuous partial derivatives. Then,

f is analytic if and only if U and V satisfy the Cauchy-Riemann equation.

Proof Suppose that f is analytic. For $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+h) - f(z)}{h} = \frac{U(x+h,y) - U(x,y)}{h} + i \frac{V(x+h,y) - V(x,y)}{h}$$

Letting $h \rightarrow 0$, then $f'(z) = \frac{\partial U}{\partial x}(x,y) + i \frac{\partial V}{\partial x}(x,y)$.

Meanwhile, for $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+ih) - f(z)}{i \cdot h} = -i \frac{U(x,y+h) - U(x,y)}{h} + i \frac{V(x,y+h) - V(x,y)}{h}$$

Letting $h \rightarrow 0$, then $f'(z) = -i \frac{\partial U}{\partial y}(x,y) + \frac{\partial V}{\partial y}(x,y)$

So, $U_x = V_y$ and $V_x = -U_y$. (*) [we using continuously differentiable.]

Conversely, U and V satisfy the condition (*), and U_x, U_y, V_x, V_y are continuous.

Let $z = x+iy \in G$ be given. Since G is open, for some $r > 0$, $B_r(z) \subseteq G$.

For $h = s+it \in B_r(0)$,

$$U(x+s, y+t) - U(x,y) = [U(x+s, y+t) - U(x, y+t)] + [U(x, y+t) - U(x,y)] \quad (*)$$

By the Mean-Value Theorem for variables s and t , respectively,

$$U(x+s, y+t) - U(x, y+t) = U_x(x+s_1, y+t) \cdot s \text{ for some } |s_1| < |s|$$

$$\text{and } U(x, y+t) - U(x,y) = U_y(x, y+t_1) \cdot t \text{ for some } |t_1| < |t|$$

Let $Q(s,t) = [U(x+s, y+t) - U(x,y)] - [U_x(x,y) \cdot s + U_y(x,y) \cdot t]$. Now, (*) gives

$$\frac{Q(s,t)}{s+it} = \frac{s}{s+it} \cdot [U_x(x+s_1, y+t) - U_x(x,y)] + \frac{t}{s+it} \cdot [U_y(x, y+t_1) - U_y(x,y)]$$

But, $|s_1| < |s| \leq |s+it|$ and $|t_1| < |t| \leq |s+it|$, so letting $s+it \rightarrow 0$, then

$$\frac{Q(s,t)}{s+it} \rightarrow 0$$

Therefore $U(x+s, y+t) - U(x,y) = U_x(x,y)s + U_y(x,y)t + Q(s,t)$

and similarly $V(x+s, y+t) - V(x,y) = V_x(x,y)s + V_y(x,y)t + R(s,t)$ where $\lim_{s+it \rightarrow 0} \frac{R(s,t)}{s+it} = 0$.

$$\text{Now, } \frac{f(z+s+it) - f(z)}{s+it} = U_x(z) + iV_x(z) + \frac{Q(s,t) + iR(s,t)}{s+it} \rightarrow U_x(z) + iV_x(z).$$

as $s+it \rightarrow 0$.

Thm. Let G be either $\mathbb{C}$ or some open disk

If $U: G \rightarrow \mathbb{R}$ is a harmonic, then U has a harmonic conjugate.

There is, there is $V: G \rightarrow \mathbb{R}$ s.t. $U + iV$ is analytic in G .

Proof. Let $G = B_R(0)$, $0 < R \leq \infty$. For a harmonic $U: G \rightarrow \mathbb{R}$, define

$$V(x, y) = \int_0^y U_x(x, t) dt + \varphi(x).$$

where $\varphi(x)$ s.t. $V_x = -U_y$. Now, differentiating both sides with respect to x :

$$V_x(x, y) = \int_0^y U_{xx}(x, t) dt + \varphi'(x)$$

$$= \int_0^y -U_{xy}(x, t) dt + \varphi'(x)$$

$$= -U_y(x, y) + U_y(x, 0) + \varphi'(x)$$

So, $\varphi'(x) = -U_y(x, 0)$. Now,

$$V(x, y) = \int_0^y U_x(x, t) dt - \int_0^x U_y(s, 0) ds$$

satisfies the Cauchy-Riemann equation.

Zeros of Analytic functions.

Def. If $f: G \rightarrow \mathbb{C}$ is analytic and $a \in G$ s.t. $f(a) = 0$,
then a is called a zero of f of multiplicity $m \geq 1$ if:

there exists an analytic $g: G \rightarrow \mathbb{C}$ s.t. $f(z) = (z-a)^m g(z)$, where $g(a) \neq 0$.

Index of a closed Curve.

Let $\gamma(t) = a + e^{2\pi i n t}$ ($0 \leq t \leq 1$). Then,

$$\int_{\gamma} \frac{1}{z-a} dz = \int_0^1 e^{-2\pi i n t} \cdot 2\pi i n \cdot e^{2\pi i n t} dt = 2\pi n \cdot i.$$

In general,

Lemma. Let $\gamma: [0,1] \rightarrow \mathbb{C}$ be a closed rectifiable curve and $a \notin \text{Im } \gamma$, then

$$\frac{1}{2\pi i} \cdot \int_{\gamma} \frac{1}{z-a} dz$$

is an 'integer'.

Proof. Assume that γ is a smooth map. Define $g: [0,1] \rightarrow \mathbb{C}$ as

$$g(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s)-a} ds$$

Then, $g(0) = 0$ and $g(1) = \int_0^1 \frac{\gamma'(s)}{\gamma(s)-a} ds = \int_{\gamma} \frac{1}{z-a} dz$

And $g(t) = \frac{\gamma(t)}{\gamma(t)-a}$ for $0 \leq t \leq 1$.

But, $\frac{d}{dt} e^{-g(t)} \cdot (\gamma(t)-a) = e^{-g(t)} \cdot \gamma'(t) - g'(t) \cdot e^{-g(t)} \cdot (\gamma(t)-a) = 0$

So, $e^{-g(t)}(\gamma(t)-a)$ is the constant,

$$e^{-g(0)}(\gamma(0)-a) = \gamma(0)-a = e^{-g(1)}(\gamma(1)-a)$$

Since γ is closed curve, so $\gamma(1) = \gamma(0)$,

$$e^{-g(1)} = 1 \quad \text{or} \quad g(1) = 2k\pi \cdot i \quad \text{for some } k \in \mathbb{Z}.$$

Def. If γ is a closed rectifiable curve in $\mathbb{C}$ and $a \notin \text{Im } \gamma$,

$$n(\gamma; a) \stackrel{\text{def}}{=} \frac{1}{2\pi i} \cdot \int_{\gamma} \frac{1}{z-a} dz$$

is called the index of γ with respect to a .

Morera Thm.

Thm. Let $f: G \rightarrow \mathbb{C}$ be a continuous function on a region G such that

$$\int_T f = 0$$

for every triangular path T in G , then f is analytic in G .

Proof. Enough to show that f has a primitive wlog, let $G = B_R(a)$. Define

$$F(z) = \int_{[\alpha, z]} f(w) dw, \text{ where } [\alpha, z]: (1-t)\alpha + tz, \quad 0 \leq t \leq 1.$$

Let $z_0 \in G$ be fixed. Then, $F(z) = \int_{[\alpha, z_0]} f + \int_{[z_0, z]} f$, so

$$\frac{F(z) - F(z_0)}{z - z_0} = \frac{1}{z - z_0} \cdot \int_{[z_0, z]} f$$

$$\begin{aligned} \text{Therefore, } \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) &= \frac{1}{z - z_0} \cdot \int_{[z_0, z]} f - f(z_0) \\ &= \frac{1}{z - z_0} \cdot \int_{[z_0, z]} [f(w) - f(z_0)] dw. \end{aligned}$$

$$\text{Now, } \left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| = \left| \frac{1}{z - z_0} \cdot \int_{[z_0, z]} [f(w) - f(z_0)] dw \right| \leq |f(z) - f(z_0)| \rightarrow 0.$$

